

Engineering a Stretch-Perfusion Bioreactor Unit to Promote Smooth Muscle Cell Growth for Urethral Tissue Reconstruction

Capstone Team K

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Abstract

This project addresses a critical gap in tissue engineering by developing a novel bioreactor capable of replicating dual mechanical properties essential for regenerating complex tissues, such as male urethral replacements. Existing bioreactors, limited to single-cell cultures with uniform properties, fail to support the differentiation of multiple distinct cell types required for layered tissue structures. The proposed bioreactor integrates synchronized axial stretching and radial perfusion, utilizing an external motor and pump-driven oscillation. This setup enables co-culture of smooth muscle and epithelial cells on a cylindrical scaffold, mimicking native urethral architecture. By engineering tissues with varied mechanical characteristics, this bioreactor advances technical capabilities, promotes functional tissue constructs, and addresses a current open niche in micro-bioreactors for regenerative medicine.

1. Introduction

1.1 Background

Current bioreactors for tissue engineering are most often limited to single-cell cultures¹ with uniform properties, which falls short for complex tissues that require multiple distinctly differentiated cell types. Differentiation of cell phenotypes occurs via mechanical, chemical, or electrical stimuli during maturation for bioreactors.² Existing designs for mechanical stimulation typically support either axial stretch or radial perfusion independently,^{3,4} limiting their ability to replicate the dynamic environment needed for complex tissue development with multiple cell phenotypes. Notably, there are no bioreactors specifically designed for urethral cells in the literature today. Furthermore, no existing bioreactors incorporate all the necessary factors—perfusion, stretch in a tubular form, and the combination of contrasting mechanical functions—to support co-culture and cell differentiation for two distinct cell types. To address this gap in tissue engineering, our micro-bioreactor introduces the novel approach of synchronizing axial stretch and radial perfusion through an external motor and oscillating pump driven flow. Axial stretch is achieved by elongating a cylindrical scaffold, while radial stretch is induced by perfusion via pump driven oscillating flow.⁵ The decellularized tissue scaffold acts as an intermediate component between the inlet and outlet flows of the pump system. Unlike other designs that restrict interior flow, our scaffold allows direct media contact, enabling outward perfusion that supports cell growth. By simulating physiological conditions required for smooth muscle and epithelial cell maturation, this microbioreactor promotes the differentiation of two distinct cell types within the same configuration.¹ The design may be customized for clinical applications in regenerative medicine for organs that rely on multiple cell types, such as heart valves or intermediate tissues.

The setup not only applies realistic mechanical stimuli but also delivers nutrients and biochemical signals directly to the cells, enhancing cellular development and reducing the risk of non-uniform growth.⁶ The application of two different types of oscillating forces represents an improvement over existing approaches in micro-bioreactor design¹ for the field as no other alternative devices exist to culture two different cell types for the express purpose of tissue engineering. The approach is particularly useful for urethral tissue engineering where structural integrity and functional layer distinction are essential.⁷ If successful, this bioreactor could be adapted for other similar applications, where complex layered tissue structures are needed. This could potentially set a new standard in clinical bioreactor design and tissue engineering.

The lack of effective urethral tissue replacements presents a significant barrier in treating congenital and acquired urethral defects, such as those seen in hypospadias,⁸ where the opening of the urethra is located on the shaft of the penis instead of the tip. Current urethral substitutes, often sourced from donor tissues like the cheek or skin, fail to replicate the native urethral architecture and function. Native urethral tissue is uniquely structured, with an inner epithelial layer composed of urothelial cells that prevent urine leakage, a middle layer of smooth muscle (**Figure 1**) and connective tissue providing structural support, and an outer vascularized corpus spongiosum layer aiding in urine propulsion.⁹ Due to these structural complexities, creating a tissue-engineered construct that can mimic these properties remains a challenge in regenerative medicine.

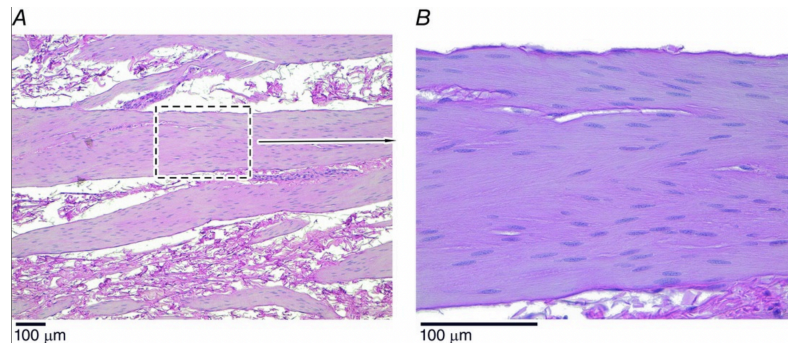


Figure 1⁷: Histological section of the proximal urethra. A) General view of muscle structures in a longitudinal section. B) Detail to highlight the inner conformation of muscle

1.2 Approach

Our approach includes engineering a bilayer construct that closely replicates the native urethral structure. This construct will feature an inner epithelial layer and an outer layer of muscle cells, grown within advanced 3D-printed and bladder acellular matrix (BAM) scaffolds designed to support cell growth and differentiation.¹⁰ Mechanical properties and cellular microenvironments play an important role in directing cell differentiation, with stiffer environments promoting the growth of more rigid tissue types and softer environments supporting softer tissues.¹¹ The harvested cells used for conventional grafting are often stiffer than urethral epithelium cells, leading to functional mismatches in the tissue construct.¹² Addressing this problem, our scaffolds are designed to provide the appropriate mechanical cues and will be supported via both aforementioned mechanisms. These capabilities are necessary for inducing both the smooth muscle phenotype necessary for the urethral outer layer and the urothelium phenotype for the inner layer.

This project faces two key challenges: creating a tubular, stretchable bioreactor scaffold system and establishing bioreactor conditions to promote cell differentiation for functional urethral tissue. Smooth muscle cells (SMCs), vital for urethral structural support, require electrochemical and mechanical cues – mimicking stretching – to achieve the desired phenotype. Success would advance tissue-engineered urethral constructs and improve clinical outcomes for urethral reconstruction, particularly for hypospadias, by enhancing tissue compatibility and functionality.

Cyclic stress testing will evaluate scaffold material properties, ensuring tensile data remains within the elastic region for non-permanent deformation during vertical stretching. Oscillation testing will determine the pump flow pressure tolerances and outward stretch felt by the scaffold. Based on these results, cells will be subjected to tunable mechanical stresses to develop the desired phenotype. Their mechanical properties will be compared to cells matured without stimuli, enabling phenotypic characterization and tuning of maturation processes to achieve optimal mechanical properties. The phenotypic characterization of the different cells will allow

for tunability in the maturation process via stress, allowing for gradual convergence toward desired mechanical properties.

2. Materials and Methods

2.1 Bioreactor Design and Components

An in depth list of our materials for our physical prototype are shown in **Table 1** under the Appendix section. Structures and design considerations, focused on material constraint and fabrication of structures, are further described in the proceeding section subparts.

2.1.1 Structural Design and Modeling

The bioreactor was designed to deliver synchronized axial stretching and radial perfusion for the cultivation of multi-layered tissue constructs. To achieve precise control over the mechanical environment, key components—including the well plate, well plate lid, scaffold support tubes, basin/linkage, I-bar, and stepper motor frame—were modeled using SolidWorks. The design emphasizes modularity, sterility, and compatibility with standard incubator environments within a closed-loop system to facilitate live cell testing under controlled mechanical conditions.

2.1.2 3D Printing and Fabrication Techniques

The bioreactor components were fabricated using a combination of Fused Deposition Modeling (FDM) and Stereolithography (SLA), with material selection based on the specific mechanical and biological requirements of each component. For components that did not come into direct contact with cell cultures, FDM 3D printing was employed using polypropylene filament. Polypropylene was selected for its autoclavability, mechanical durability, and relative inertness, ensuring that parts such as the I-bar, basin, and stepper motor frame could withstand repeated sterilization cycles without degradation.

In contrast, components that directly interacted with the cell culture environment were fabricated using SLA 3D printing with *Liqcreate Bio-Med Clear* resin. This resin was chosen due to the inherent biocompatibility, autoclavability, and certification under **ISO 10993-5:2009**, confirming its compliance with standards for assessing the in vitro cytotoxicity of medical devices. The well plate, well plate lid, scaffold tubing supports, and tubing connector were all produced using this material to ensure a sterile and non-toxic environment for cell culture.

2.1.3 Sterilization and Assembly Protocol

Maintaining a sterile environment throughout the fabrication and assembly process was critical in ensuring the viability of the cultured tissues. To achieve this, all components underwent rigorous sterilization protocols tailored to the materials used. Components fabricated from polypropylene and *Bio-Med Clear* resin, as well as silicone tubing and metal hardware (clamps, linear guides, screw shafts, screws, and nuts), were sterilized via an autoclave. The process ensured the elimination of potential contaminants while preserving the structural integrity of the components. For components that could not withstand autoclaving, such as motors and electronic control units, sterilization was achieved by wiping down surfaces with 70% isopropyl alcohol (IPA). This method minimized contamination risks while maintaining the functionality of sensitive electronic parts.

The initial assembly of the bioreactor was conducted in a biosafety cabinet to maintain sterility. Once assembled, the bioreactor was transferred into a 37°C, 5% CO₂ incubator for live cell testing. The Arduino-based control system and peristaltic pump were positioned externally to minimize contamination risks while enabling precise mechanical stimulation.

2.1.4 Scaffold Attachment and Operational Set-up

The attachment of the seeded scaffold to the bioreactor was performed in two stages to minimize handling time and reduce contamination risk. First, within a biosafety cabinet, the biomaterial scaffold was placed onto a custom well plate, preloaded with inner tubing supports, and secured with a stainless steel 304 clamp. This step was completed in under five minutes to ensure that the scaffold remained in a sterile environment. The assembled scaffold was then transferred into the incubator, where it was connected to the perfusion tubing and secured to the support stand, a process requiring an additional one to two minutes. This streamlined setup facilitated the rapid initiation of mechanical stimulation while maintaining sterile conditions.

2.1.5 Design Improvements and Functional Considerations

The bioreactor design incorporated several improvements to address the limitations encountered in early prototypes. One of the primary challenges was scaffold tearing under mechanical stress, which was mitigated by integrating *Bio-Med Clear* resin structures. These structures were specifically designed to evenly distribute mechanical forces across the scaffold, reducing stress concentrations during axial stretching and radial perfusion.

Another key improvement focused on media containment. The scaffold housing was engineered to be liquid-tight, allowing for continuous perfusion without leakage or net media loss. This was achieved with the use of special bellows made of FDA approved and autoclavable silicon. These compliant mechanisms allowed the stretching motion from the motor to translate into the well plates with the cells while ensuring a watertight seal. The media system ensured a consistent supply of nutrients and waste removal, both of which are essential for cell viability and healthy tissue maturation. Furthermore, the bioreactor was optimized to support live cell stimulation in a sterile environment. The combination of autoclavable materials, external control systems, and streamlined assembly protocols allowed for precise mechanical stimulation while minimizing contamination risks. This comprehensive approach enables the bioreactor to support complex multi-layered tissue engineering applications, including urethral reconstruction and potentially other tubular tissue models.

2.2 Computational Fluid Dynamics (CFD)

COMSOL Multiphysics was used to simulate media flow and cyclic stretching in the scaffold, employing a combination of laminar flow and solid mechanics modules to capture the central perfusion and stretch mechanisms of the bioreactor. The scaffold is modeled using a hyperelastic material model, and accurate flow behavior over time is simulated through the application of fluid-structure interaction between the flow domain and surrounding solid bodies, in a time-dependent study.

2.3 Scaffold Preparation

For this study, decellularized rabbit urethral tissues, or bladder acellular matrix (BAM), were employed as scaffolds. These tissues were previously decellularized in our laboratory using a standardized protocol designed to preserve the extracellular matrix (ECM) while effectively

removing immunogenic cellular components. The decellularized scaffolds provided a structurally stable and biologically relevant matrix for cell seeding, supporting the architectural requirements necessary for bioreactor testing.

Following decellularization, the rabbit urethral scaffolds were stored in sterile phosphate-buffered saline (PBS) at 4°C until use. Prior to cell seeding, the scaffolds were equilibrated in Dulbecco's Modified Eagle Medium (DMEM) at 37°C for 1 hour to promote hydration of the ECM and optimize conditions for subsequent cell attachment.

2.3.1 Scaffold Durability Testing

After hydration, the BAM scaffold was connected to the inlet and outlet ends of the tubing connected to the peristaltic pump, and further constricted with the SS 304 clamps. The motor was adjusted to apply a 10% stretch based on the intermediate scaffold length of 2 cm (therefore having a stretch of 2 mm applied via the linear actuator). The frequency was set to 2 Hz and allowed to run for 48 hours which translates to 85,000+ cycles of stretch on the scaffold.

3. Results and Discussion

3.1 Bioreactor Assembly

To verify the design's functionality, an initial benchtop prototype was built to validate performance before introducing biological materials. The prototype met our minimum design criteria: a bioreactor capable of growing co-culture cells on a tubular scaffold with perfusion and oscillating stretch mechanisms. We printed biomedical scaffold supports, secured the stepper motor, connected the screw shaft, and integrated the media perfusion system to assess flow rate and structural integrity.

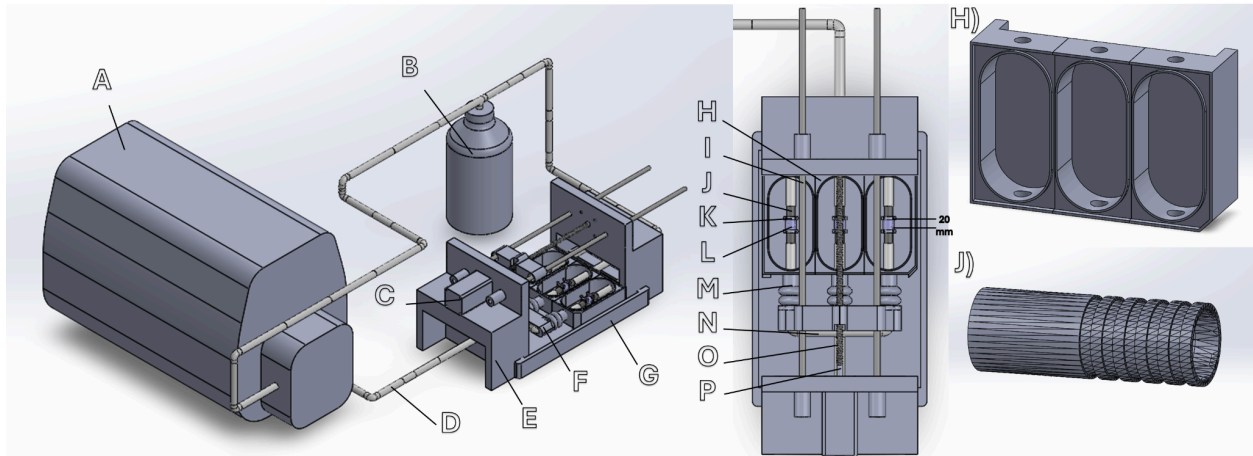


Figure 2: Complete Freestanding Bioreactor Configuration

The current prototype consists of a peristaltic pump (A) that supplies cell media through silicone tubing (D) from a media bottle (B) to the cell scaffold (L). The scaffold is secured to the scaffold support tubes (J) using 304 stainless steel clamps (K) and housed in a custom 3-well plate (H). Furthermore, a stepper motor (C), mounted on a stepper motor frame (E), drives a screw shaft (O) connected to an I-bar (F) on one side and a coupler (P) on the other, moving along linear guides (I) to apply a controlled linear stretch to the cell scaffolds. Additionally, a 3-to-1 tubing connector (N) distributes media from the peristaltic pump to each scaffold through silicone bellows (M), which prevent leaks by oscillating with the tubing. The well plate (H) is mounted on the basin (G), which connects to the stepper motor frames, ensuring system stability relative to scaffold stretching.

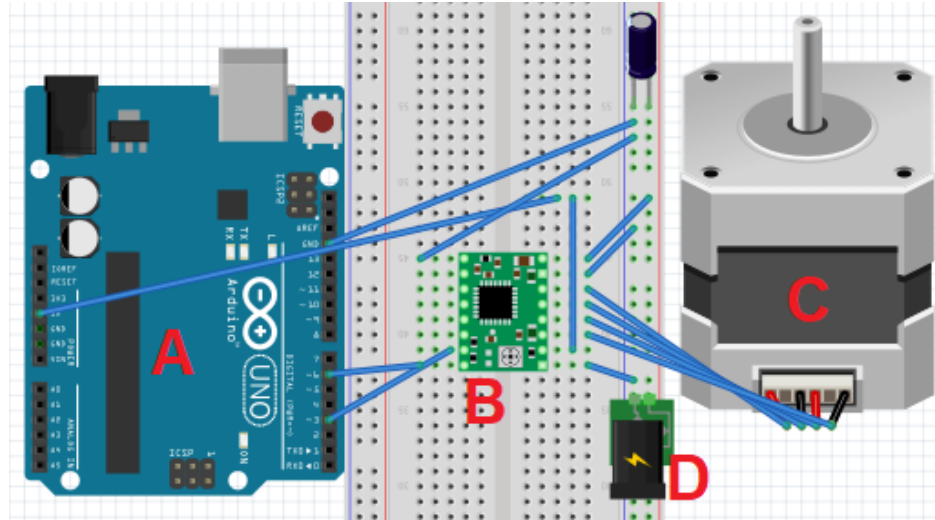


Figure 3: Electronic Control Circuit

This figure shows the circuit controlling the linear stretch mechanism. An Arduino microcontroller (A) signals a stepper motor controller (B), which drives a Nema-11 stepper motor (C), connected to the screw shaft. Power is provided via a transformer, connected to the circuit at point D.

By testing and identifying issues such as scaffold tearing, media flow inconsistencies, and sterility challenges, we further refined our bioreactor design. The revised prototype was scaled for incubator compatibility and live cell testing, integrating synchronized axial and radial stretching, improved sterility management, and solutions to earlier limitations (**Figure 2**). A simple Arduino circuit was used to control the linear actuator and apply the proper amount of stretching to the cell scaffold (**Figure 3**).

This design choice was informed by the mechanical properties of the BAM scaffolds, which exhibit low ultimate tensile strength (UTS) and Young's Modulus (528 kPa), making them prone to deformation and failure under stress. SS 304 clamps were also incorporated to secure the scaffold during operation, ensuring stability during stretching.

When testing our bioreactor with live cells for the first time, more issues became apparent in the implementation process. The setup had to be assembled inside of a biosafety cabinet in order to maintain sterility. Furthermore, our scaffolds were sutured to the incorrect inner diameter, resulting in incompatibility with our scaffold support tubes. To counteract this, we modified the bioreactor to be assembled on a workbench, and then sterilized in one piece by employing autoclavable 3D printing filament (polypropylene). We also confirmed with the cell suturing team the correct dimensions in order to avoid having our future scaffolds being sutured to the wrong dimensions.

3.2 CFD Simulations

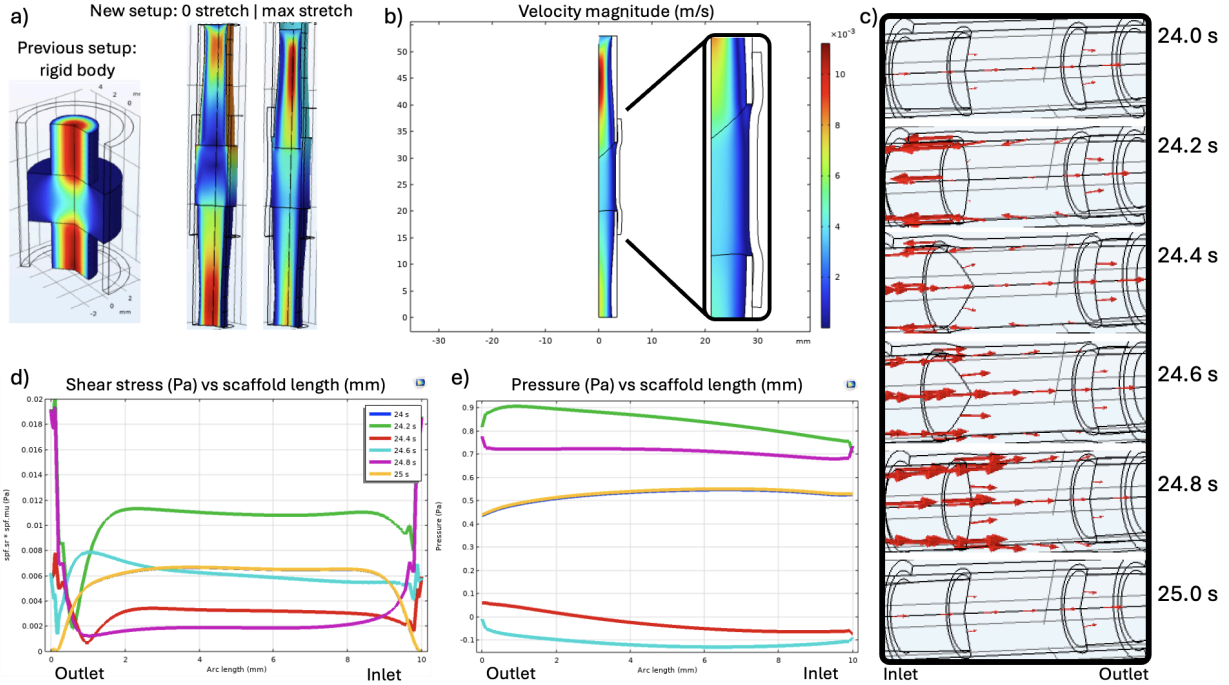


Figure 4: Velocity Profiles, Pressure, and Shear Stress (BAM) at 5 mL/min, 15% Stretch

(A) Evolution from initial rigid body study setup to final cyclic stretch + perfusion setup. (B) Velocity flow showcasing scaffold deformation. (C) Velocity streamlines and direction through scaffold during one stretch cycle. (D) Changing WSS along scaffold for one stretch cycle. (E) Pressure drop across scaffold-support tube assembly for one stretch cycle. Legend in part (D) corresponds to data in part (E) as well.

CFD simulations were run in COMSOL Multiphysics and initially modeled the scaffold and connecting supports as a rigid body (**Figure 4a**), to gain a rough overview of the wall shear stress (WSS) and flow velocity that would be experienced by the scaffold. At a flow rate of 5 mL/min, the maximum wall shear stress experienced by the (rigid) scaffold was 0.11 dyne/cm^2 , within the range of expected values ($0.1 - 1 \text{ dyne/cm}^2$) determined from literature review.¹⁵ Subsequent simulations implemented fluid-structure interaction to account for the stretchability of the scaffold and model the system during the perfusion mode of the bioreactor. Due to the unique material composition of BAM (the scaffold mimic used for experiments), a custom material was created in COMSOL (elastic modulus: 528 kPa, Poisson's ratio: 0.45, density: 1200 kg/m^3). Water was used to model the media flow through the scaffold and scaffold supports.

For modeling the conditions during the combined stretch-perfusion mode, studies were performed inducing 0% stretch (control), 10% cyclic stretch, and 15% cyclic stretch on the scaffold at various flow rates and at 1 Hz.⁶ The five flow rates tested were 1 mL/min, 5 mL/min, 10 mL/min, 20 mL/min, and 100 mL/min. **Figure 4b** shows scaffold deformation due to 15% stretch applied at the top support, which is moved upward from its normal position. Cyclic stretch induced non-uniform flows along the fluid-scaffold interface, with the motion of the support tube causing frequent directional changes (**Figure 4c**). WSS along the scaffold similarly changes rapidly within every stretch cycle (**Figure 4d**), ranging approximately from 0.02 to 0.11 dyne/cm^2 for an inlet flow rate of 5 mL/min during this period. The pressure drop across the whole scaffold-support

tube assembly exhibits directional oscillation within each stretch cycle. **Figure 4e** shows that the pressure drop across the scaffold is very small; not exceeding 1 Pa.

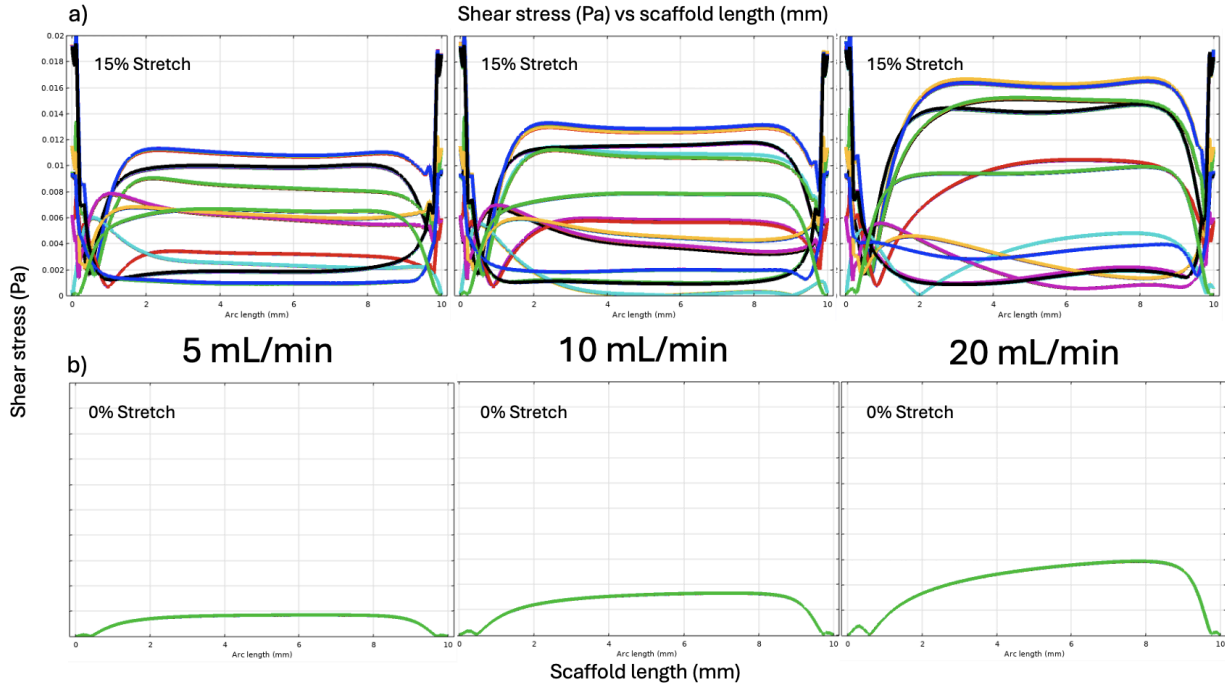


Figure 5: Irregular Wall Shear Stress along Scaffold at Higher Flow Rates

(A) Comparison of shear stress along scaffold with 15% stretch applied, flow rates 5 through 20 mL/min . (B) Comparison of shear stress along scaffold with 0% stretch applied, flow rates 5 through 20 mL/min . In both (A) and (B), shear stress along the scaffold becomes more irregular at higher flow rates. All graphs range from 0 - 10 mm in x-axis, from 0 - 0.02 Pa (or 0 - 0.2 dyne/cm^2) in y-axis. Each individual line corresponds to one time point, range: 20 - 25 s.

To confirm which range of flow rates would be ideal for our application, we studied the changes in shear stress along scaffold length with various flow rates. As seen in **Figure 5**, shear stress along the scaffold becomes more irregular with higher flow rates, regardless of if stretch is applied or not. This informed our decision to narrow down the range of experimental flow rates to between 5 and 10 mL/min . Observing the graphs for 15% stretch, it is clear that both 5 and 10 mL/min flow rates are sufficient to achieve our minimum desired shear stress of 0.1 dyne/cm^2 .

When observing real experimental conditions, which will be touched upon further in section 3.3, we observed that the actual BAM scaffold has a slightly smaller diameter than modeled in CFD, due to the fact that it is sutured tightly around a mold during tubularization. This results in slightly larger real shear stress values than those predicted by computational modeling.

These data showcase how, during both perfusion-only and stretch-perfusion conditions, a flow rate of 5 mL/min and 15% stretch applied to each experimental replicate is sufficient to achieve adequate shear stresses for inducing cell differentiation.

3.3 Benchtop Testing

To validate the functionality of the bioreactor system, a series of benchtop tests were conducted to assess the perfusion, axial stretching, and full system integration before introducing biological

scaffolds. These tests ensured that the system components operated effectively and reliably under experimental conditions.

3.3.1. Perfusion Testing

The first phase of benchtop testing focused on confirming the media perfusion mechanism. After calibrating the peristaltic pump, dyed water was pumped through the inner tubing of the scaffold, simulating nutrient delivery in a dynamic culture environment. As shown in **Figure 6**, the dyed water successfully traveled from the peristaltic pump, through the silicone tubing and cylindrical resin supports, radially stretching the scaffold before exiting through the outflow tubing into the waste media reservoir. This test confirmed that the media flow pathway was unobstructed, and that perfusion could be sustained throughout the experiment.

3.3.2. Axial Stretching Validation

Next, the stretch-mechanism was evaluated independently to assess whether the stepper motor and linear actuation system could reliably generate controlled axial strain on the scaffold. As shown in **Figure 7**, the stepper motor was successfully connected to the bioreactor tubing via a 3D-printed fixture, allowing for controlled oscillatory motion. The mechanism was tested across multiple cycles, demonstrating consistent stretching without mechanical failure, confirming the feasibility of the axial actuation system.

3.3.3. Full System Integration with a Silicone Scaffold

Following individual component validation, a silicone scaffold was used to test the fully assembled bioreactor system, integrating both perfusion and stretching mechanisms. As shown in **Figure 8**, the silicone scaffold successfully deformed in response to axial stretching, while perfusion was maintained throughout the cycle. This test confirmed the compatibility of the two systems, demonstrating that the scaffold could undergo both radial expansion and axial stretch in a synchronized manner, simulating the expected conditions for biological cultures.

3.3.4. High Cycle Fatigue Testing with BAM Scaffold

Once the system was validated using a silicone scaffold, testing was conducted with the bladder acellular matrix (BAM) scaffold to assess long-term mechanical performance. The BAM scaffold was subjected to high cycle fatigue testing as shown in **Figure 9**, evaluating cyclic fatigue resistance under repetitive stretching and prolonged media exposure, media containment over multiple days to ensure the well plate compartments maintained stable fluid levels, and motor performance and durability over extended operation.

The BAM scaffold successfully withstood 85,000+ cycles of mechanical stretching without signs of failure. Media levels remained stable throughout the testing period, and the stepper motor and perfusion system continued functioning without interruption, confirming the system's long-term tolerability for extended culture durations.

These benchtop tests collectively validated the bioreactor's mechanical performance, reliability, and suitability for live cell culture, ensuring a robust platform for dynamic tissue engineering experiments.

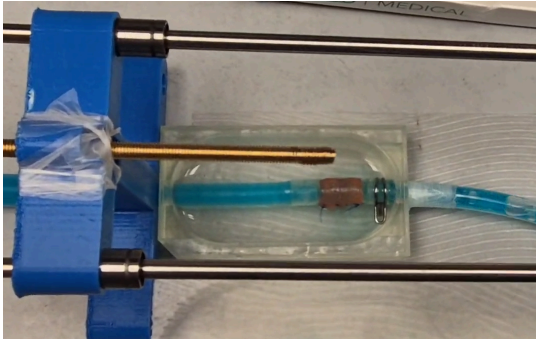


Figure 6: Perfusion Component

This figure highlights the success of our perfusion component of our benchtop bioreactor successfully demonstrating flow through our mock scaffold

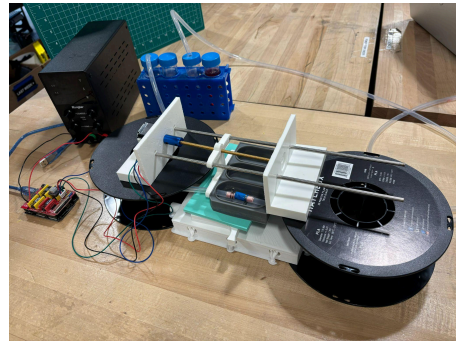


Figure 7: Stretch-Mechanism Component

This figure shows our stretch mechanism component of our benchtop bioreactor

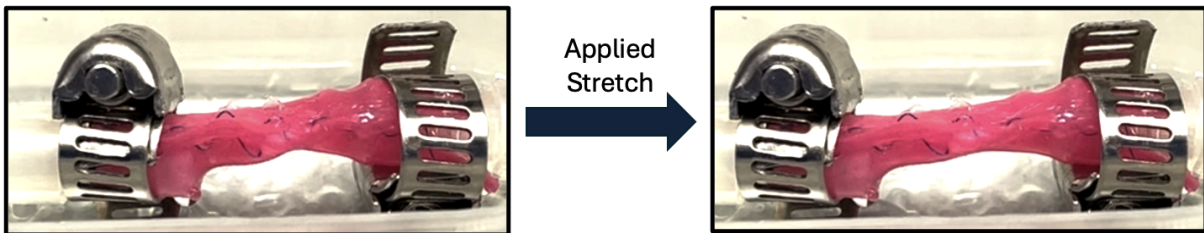


Figure 8: Scaffold Deformation with Full System Integration

This figure indicates the before and after of the stretch-mechanism component of our benchtop bioreactor in action, inducing about 10% stretch on the mock scaffold

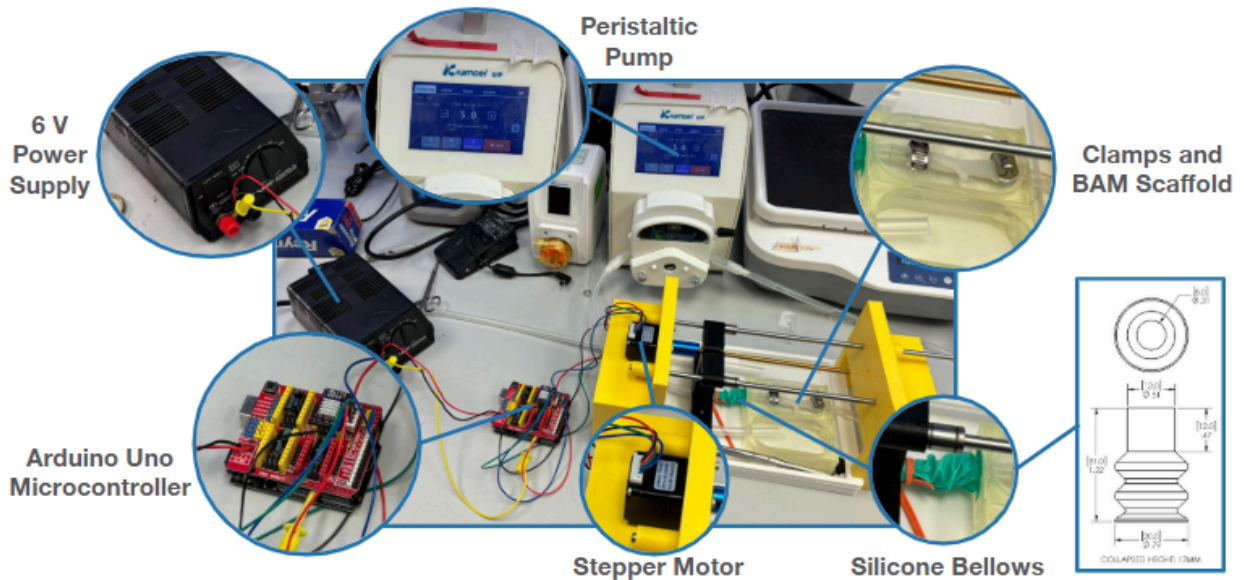


Figure 9: Complete Freestanding System, Integrating Perfusion and Stretch Components

This figure highlights the main components of our bioreactors. The 6V power supply, arduino uno microcontroller, and stepper motor are used in the stretch-mechanism component, while the peristaltic pump and silicone bellows are used in the perfusion component.

4. Conclusion

4.1 Project Design and Purpose

Our proposed bioreactor design offers a unique approach to mitigating current issues with urethral tissue replacements, which involve insufficient mechanical similarity to native urethral tissue, by closely mimicking the layered urethral tissue structure. Our bioreactor applies both radial perfusion and axial oscillating stretch to a cell-seeded scaffold, thereby inducing cells seeded on the outside of the scaffold to differentiate into smooth muscle cells, and cells experiencing perfusion on the inside of the scaffold to differentiate into urothelial cells. The design functionality was validated using CFD and benchtop experiments to ensure it works as intended.

4.2 Future Directions

Future directions involve conducting live cell testing. Future live cell tests will involve testing different applied mechanical stresses and observing resulting phenotypes. Comparisons with desired mechanical properties will allow us to converge on a set of parameters ideal for mimicking the structures found in native urethral tissues. This is possible due to the tunable nature of the mechanisms implemented, and require both theoretical simulations and experimental testing for parameter convergence. Additional future directions and implications involve the use of the bioreactor for other tissue types, and modifying the coupling mechanism to allow for non-tubular and alternate geometries on the device.

5. Appendix

Table 1: Materials, Supplies, and Equipment

Materials	Applications
Tubing, Peristaltic Pump, Environment Compatible Connectors	Media Perfusion System
Stepper Motor, Screw Shaft, and Linear Guide Rods, Transformer, Wires, Couplings Coupler Connector	Stretch System
Liqcreate Bio-Med Clear Resin	Used with Original Prusa SL1S Speed 3D Printer. Biocompatible (per ISO 10993-5:2009) and Autoclavable
Polypropylene Filament	Used with Original Prusa MK4 3D Printer. Biocompatible and Autoclavable
Original Prusa SL1S Speed 3D Printer	Stereolithography (SLA) Resin Printer used to print Well Plate, Well Plate Lid, Tubing Connector, and Scaffold Support Tubes
Original Prusa MK4 3D Printer	Fused Deposition Modeling (FDM) 3D Printer used to print Basin, I-Bar, and Stepper Motor Frame
Software (COMSOL Multiphysics, SOLIDWORKS, Fusion360)	Modeling, Designing, and Simulating Components and Systems
Arduino, Circuits, Motor Driver	Electronics for Operating Stepper Motor

Citations

1. Grün, Christoph, Brigitte Altmann, and Eric Gottwald. 2020. "Advanced 3D Cell Culture Techniques in Micro-Bioreactors, Part I: A Systematic Analysis of the Literature Published between 2000 and 2020." *Processes* 8 (12): 1656. <https://doi.org/10.3390/pr8121656>.
2. Lee, Pei-Yuan, Yen-Ching Liu, Mei-Xuan Wang, and Jin-Jia Hu. 2018. "Fibroblast-Seeded Collagen Gels in Response to Dynamic Equibiaxial Mechanical Stimuli: A Biomechanical Study." *Journal of Biomechanics* 78 (July 30): 134–42. <https://doi.org/10.1016/j.jbiomech.2018.07.030>.
3. Talò, G., D'Arrigo, D., Lorenzi, S., Moretti, M., and Lovati, A.B. 2020. "Independent, Controllable Stretch-Perfusion Bioreactor Chambers to Functionalize Cell-Seeded Decellularized Tendons." *Annals of Biomedical Engineering* 48 (3): 1112–1126. <https://doi.org/10.1007/s10439-019-02257-6>.
4. Lei, Y., and Z. Ferdous. 2016. "Design Considerations and Challenges for Mechanical Stretch Bioreactors in Tissue Engineering." *Biotechnology Progress* 32 (3): 543-53. <https://doi.org/10.1002/btpr.2256>.
5. Syedain ZH, Tranquillo RT. Controlled cyclic stretch bioreactor for tissue-engineered heart valves. *Biomaterials*. 2009 Sep;30(25):4078-84. doi: 10.1016/j.biomaterials.2009.04.027. Epub 2009 May 26. PMID: 19473698; PMCID: PMC2762550.
6. Soroor Sharifpoor, Craig A. Simmons, Rosalind S. Labow, J. Paul Santerre, "Functional characterization of human coronary artery smooth muscle cells under cyclic mechanical strain in a degradable polyurethane scaffold, *Biomaterials*, Volume 32, Issue 21, 2011, Pages 4816–4829, ISSN 0142-9612, <https://doi.org/10.1016/j.biomaterials.2011.03.034>.
7. Natali, A.N., Carniel, E.L., Frigo, A., Pavan, P.G., Todros, S., Pachera, P., Fontanella, C.G., Rubini, A., Cavicchioli, L., Avital, Y., and De Benedictis, G.M. 2016. "Experimental Investigation of the Biomechanics of Urethral Tissues and Structures." *Experimental Physiology* 101: 641-656. <https://doi.org/10.1113/EP085476>.
8. Halaseh SA, Halaseh S, Ashour M. 2022. "Hypospadias: A Comprehensive Review Including Its Embryology, Etiology and Surgical Techniques." *Cureus*. <https://doi.org/10.7759/cureus.27544>.
9. Wein, Alan J., Louis R. Kavoussi, and Andrew C. Novick, eds. 2015. "Anatomy of the Lower Urinary Tract and Male Genitalia." In *Campbell-Walsh Urology*, 11th ed. Philadelphia: Elsevier.
10. Jha, Amit, and Debopriya Ray. 2021. "The Role of Mechanical Properties of Scaffold in Tissue Engineering." *International Journal of Biomaterials*, Article ID 6678667. <https://doi.org/10.1155/2021/6678667>.
11. Place, Laura W., Theresa A. George, and Kacey L. Williams. 2020. "Engineering 3D-Printed Scaffolds for Tissue Engineering: A Review on Composition, Bioprinting Methods, and Biomechanics." *Journal of Biomedical Materials Research Part A* 108 (11): 2459-2472. <https://doi.org/10.1002/jbm.a.37053>.
12. Badylak, Stephen F., and Thomas W. Gilbert. 2008. "Immune Response to Scaffold Biomaterials." *Seminars in Immunology* 20 (2): 109-116. <https://doi.org/10.1016/j.smim.2007.11.003>.
13. Benign Urology Laboratory. 2024. Tensile Data for Human Tissue and Bioinks, Slides 1-5. Department of Urology, University of California, Los Angeles.
14. Egger, D., Fischer, M., Clementi, A., Ribitsch, V., Hansmann, J., and Kasper, C. 2017. "Development and Characterization of a Parallelizable Perfusion Bioreactor for 3D Cell Culture." *Bioengineering* 4 (2): 51. <https://doi.org/10.3390/bioengineering4020051>.
15. Zalewska-Piątek, Beata et al. "A shear stress micromodel of urinary tract infection by the *Escherichia coli* producing Dr adhesin." *PLoS pathogens* vol. 16,1 e1008247. 9 Jan. 2020, doi:10.1371/journal.ppat.1008247